

3.68 $f(x) = x^k$

Prob. 3.68 este echivalentă cu prob.

$$\min \left\{ \int_{-1}^1 [f(t) - q(t)]^2 \frac{dt}{\sqrt{1-t^2}}, q \in P_{k-1} \right\} \text{ pt } k=1, \dots, 5$$

Aproximăm pe q cu polinoame Chebishev de speța I

$\Rightarrow$ Sol. problemei este

$$q^1(t) = \frac{c_0}{2} + c_1 T_1(t) + \dots + c_{k-1} T_{k-1}(t)$$

unde $c_j = \frac{2}{\pi} (f(t), T_j(t))$

$$c_j = \frac{2}{\pi} \int_{-1}^1 t^k \cos(j \arccos t) \frac{dt}{\sqrt{1-t^2}}, \quad \forall j=0, \dots, k-1$$

$$\begin{aligned} \arccos t = x & \quad t = -1 \Rightarrow x = \pi \\ & \quad t = 1 \Rightarrow x = 0 \end{aligned}$$

$$-\frac{1}{\sqrt{1-t^2}} dt = dx$$

$$\Rightarrow \boxed{c_j = \frac{2}{\pi} \int_0^\pi (\cos x)^k \cos(jx) dx}$$

k=1 $f(x) = x$, $q^1(x) = \frac{c_0}{2}$

$$c_0 = \frac{2}{\pi} \int_0^\pi \cos x \cdot \cos(0) dx = \frac{2}{\pi} \sin x \Big|_0^\pi = 0$$

$$\Rightarrow c_0 = 0 \Rightarrow \boxed{q^1(x) \equiv 0} \quad \left(\text{în mathe } c_0 = 9,71 \cdot 10^{-17} \text{ foarte aproape de } 0 \right)$$

k=2 $f(x) = x^2$, $q^1(x) = \frac{c_0}{2} + c_1 T_1(x)$

$$c_0 = \frac{2}{\pi} \int_0^\pi (\cos x)^2 \cos(0) dx = \frac{2}{\pi} \int_0^\pi \frac{\cos 2x + 1}{2} dx$$

$$= \frac{1}{\pi} \left[\frac{\sin 2x}{2} \Big|_0^\pi + \frac{1}{1} x \Big|_0^\pi \right] = 0 + \frac{1}{\pi} \cdot \pi - 0 = 1 = \boxed{c_0 = 1}$$

do bel si in Mathe.

$$c_1 = \frac{2}{\pi} \int_0^{\pi} \cos^2 t \cdot \cos t dt = \frac{2}{\pi} \int_0^{\pi} \cos^3 t dt = 0 \Rightarrow \boxed{c_1 = 0}$$

$$\Rightarrow \varphi^1(x) = \frac{1}{2}$$

$$\boxed{k=3} \quad f(x) = x^3, \quad \varphi^1(x) = \frac{c_0}{2} + c_1 T_1(x) + c_2 T_2(x)$$

$$c_0 = \frac{2}{\pi} \int_0^{\pi} \cos^3 x dx = 0$$

$$c_1 = \frac{2}{\pi} \int_0^{\pi} \cos^3 x \cdot \cos x dx = \frac{2}{\pi} \int_0^{\pi} \cos^4 x dx = \frac{2}{\pi} \cdot \frac{3\pi}{8} = \frac{6}{8} =$$

$$c_1 = 0.75 \pi$$

$$c_2 = \frac{2}{\pi} \int_0^{\pi} \cos^3 x \cdot \cos 2x dx = \frac{2}{\pi} \int_0^{\pi} \cos^3 x (2\cos^2 x - 1) dx$$

$$= \frac{4}{\pi} \int_0^{\pi} \cos^5 x dx - \frac{2}{\pi} \int_0^{\pi} \cos^3 x dx = 0 - 0 = 0$$

$$\Rightarrow \boxed{\varphi^1(x) = 0.75 T_1(x) = 0.75 \cdot x}$$

$$\boxed{k=4} \quad f(x) = x^4, \quad \varphi^1(x) = \frac{c_0}{2} + c_1 T_1(x) + c_2 T_2(x) + c_3 T_3(x)$$

$$c_0 = \frac{2}{\pi} \int_0^{\pi} \cos^4 x dx = 0.75$$

$$c_1 = \frac{2}{\pi} \int_0^{\pi} \cos^4 x \cdot \cos x dx = \frac{2}{\pi} \int_0^{\pi} \cos^5 x dx = 0$$

$$c_2 = \frac{2}{\pi} \int_0^{\pi} \cos^4 x \cos 2x dx = \frac{4}{\pi} \int_0^{\pi} \cos^6 x dx - \frac{2}{\pi} \int_0^{\pi} \cos^4 x dx$$

$$= \frac{4^2}{\pi} \cdot \frac{5\pi}{168} - \frac{2}{\pi} \cdot \frac{3\pi}{8} = \frac{10}{8} - \frac{6}{8} = \frac{4}{8} = \frac{1}{2} = 0.500$$

$$c_3 = \frac{2}{\pi} \int_0^{\pi} \cos^4 x \cos 3x dx = 0$$

$$\varphi^1(x) = \frac{0.75}{2} + \frac{1}{2} T_2(x) = \frac{0.75}{2} + \frac{1}{2} (2x^2 - 1) =$$

↖ (am folosit Wolfram alpha)

$$= x^2 + \frac{0.25}{2} = x^2 + \frac{1}{8}$$

$$\boxed{k=5}$$

Pt $k=5$ Se procedează la fel ...

Dacă vrei poți să faci calculul pt $k=5$ dar eu nu văd rostul.